SecuriAgent AI

Technical Assessment Upgrades & Verification Report

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Role: AI/ML Developer Intern
Target: Kapstone Technological Services
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1. EXECUTIVE SUMMARY

This document reports on the system enhancements executed for **SecuriAgent AI** to fulfill updated assessment directives. The platform now incorporates genuine machine learning training and test split evaluation, active sandbox threat mitigation, regulatory compliance mapping, and a multi-tab analytical user interface.

0.963	0.925	0.944	100%
PRECISION (UNSEEN TEST)	RECALL (UNSEEN TEST)	F1-SCORE METRICS	SANDBOX EXECUTION

2. TECHNICAL UPGRADES SUMMARY

Component	Previous Implementation	Upgraded Implementation
ML Training & Evaluation	Isolation Forest trained without test split or logged metrics.	Added 70/30 train-test split, genuine precision/recall/F1 metrics export to `models/evaluation_metrics.json`.
Sandbox Threat Mitigation	Static textual recommendations in compliance outputs.	Functional firewall adapter (`src_mitigation.py`) executing real IP blocks to `sandbox/firewall_rules.json`.
Dashboard UI Architecture	Single stacked layout page with basic log summaries.	Structured 3-tab Streamlit dashboard: Live Analysis, Model Metrics, and Firewall Sandbox Monitor.
API & Security Reporting	Basic telemetry response endpoints.	Asynchronous FastAPI ingestion and automated executive Markdown report generation.

3. REGULATORY COMPLIANCE & MITIGATION MAPPING

Detected Threat	NIST CSF Control	ISO 27001 Reference	Automated / Recommended Action
LOGIN_FAILURE	PR.AC-1 (Access Control)	A.9.4.2 (Secure Log-on)	Enforce account lockout policies after 5 failed authentication attempts.
PORT_SCAN	DE.CM-1 (Network Monitoring)	A.13.1.1 (Network Controls)	Execute sandbox firewall drop rule (`DROP_ALL_TRAFFIC`) on source IP.
SQL_INJECTION	PR.DS-5 (Data Protection)	A.14.1.2 (App Security)	Deploy parameterized query rules and active Web Application Firewall (WAF) filters.

4. ACTIVE SANDBOX MITIGATION ARCHITECTURE

🛡️ Automated Firewall Block Execution Flow

When an anomalous event (such as brute-force authentication failures or unauthorized scanning) exceeds risk thresholds, the operator or automated pipeline triggers `src_mitigation.py`. The adapter serializes the blocked entity into `sandbox/firewall_rules.json` with entry timestamps, policy flags, and execution status, preventing duplicate entries and providing a transparent audit trail.

5. ASSESSMENT REQUIREMENTS VERIFICATION

Requirement Task	Verification Evidence	Status
1. Genuine ML Model Training	Model trained on synthetic SOC logs (`src_model.py`) and evaluated on unseen test data with serialized metrics.	PASSED
2. Dynamic Threat Detection	Logs processed via saved `isolation_forest.pkl` pipeline rather than static UI logic.	PASSED
3. Regulatory Compliance Mapping	Dynamic lookup translating telemetry into NIST CSF and ISO 27001 control mappings.	PASSED
4. AI Sandbox Mitigation	Simulated firewall adapter executing real state modifications in `sandbox/firewall_rules.json`.	PASSED
5. API & Executive Reporting	FastAPI backend operational; downloadable Markdown incident report generator in Streamlit.	PASSED

6. CONCLUSION & DEMONSTRATION READINESS

The upgraded **SecuriAgent AI** platform fulfills all technical criteria established by Kapstone Technological Services. The repository contains full source code, saved model artifacts, evaluation logs, and sandbox audit trails. The application is completely prepared for live demonstration and repository submission ahead of the October 22, 2026 deadline.